

St George Girls High School

Year 11

End of Preliminary Course Examination

2008



Mathematics

Time Allowed: 3 hours
(plus 5 minutes reading time)

Instructions

1. Attempt all 7 questions.
2. All questions are of equal value.
2. All necessary working must be shown.
3. Begin each question on a new page.
4. Marks will be deducted for careless work or poorly presented solutions.

Question 1 – (15 marks)

Marks

- a) Calculate to three significant figures $\sqrt{\frac{13.5^2 - 4.3^3}{29.2 \times 15.4}}$ 1
- b) Expand and simplify $(3 - 2x)^2 - 4(3x + 5)$ 2
- c) Solve $2x^2 - 15 \geq -7x$. 2
- d) Rationalise the denominator of $\frac{7 + 4\sqrt{3}}{7 - 4\sqrt{3}}$ 2
- e) Simplify $\cot \theta \times \sin \theta$ 1
- f) What is the exact value of $\operatorname{cosec} 240^\circ$? 2
- g) Find the derivatives of: 5
- (i) $y = 2\sqrt{x}$
- (ii) $y = (3x - 4)^4$
- (iii) $y = \frac{2x - 1}{2x + 1}$

Question 2 – (15 marks)

Marks

a) Simplify $\frac{2x}{x-3} + \frac{3x}{x-2}$

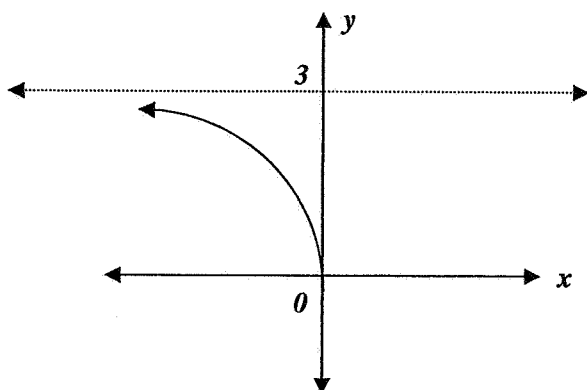
2

b) Find the equation of the tangent to the curve $y = 3x^2 - 2x + 1$ at the point $P(-1, 6)$

3

c) Using the graph drawn below.

4



(i) Copy the graph onto your answer booklet and complete the graph if it is known to be odd.

(ii) Write down the domain and range of the completed function.

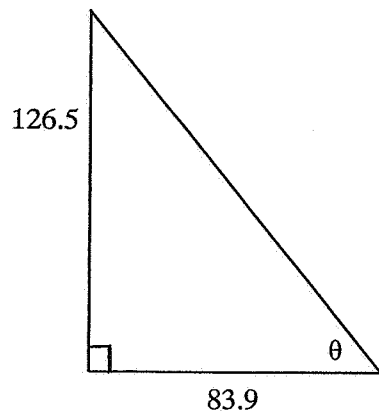
Question 2 (cont'd)

Marks

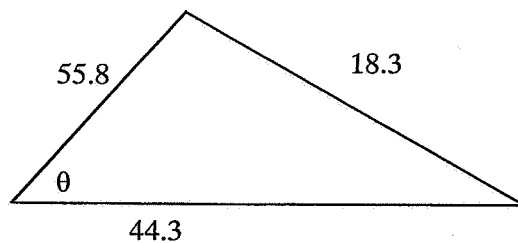
d) Calculate θ correct to the nearest minute.

4

(i)



(ii)



e) Simplify $14x^{-2}y^{-1} \div 7^2xy^{-3}$ giving the answer in index form without negative indices.

2

Question 3 – (15 marks)

Marks

a) Show that the points $A(1, -1)$, $B(3, 5)$ and $C(4, 8)$ are collinear.

3

b) Using the sequence 8, 11, 14 ... , find

5

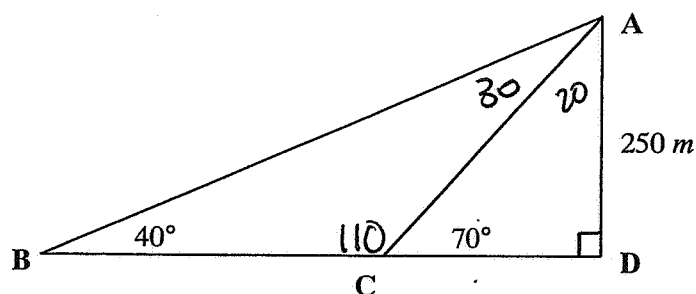
(i) the formula for the n th term in its simplest form.

(ii) the first term which is greater than 1000.

(iii) the value of the sum to twenty terms.

c) Given the following diagram.

7



(i) Show that $AC = \frac{250}{\sin 70^\circ}$

(ii) Prove that $\hat{BAC} = 30^\circ$

(iii) Calculate BC correct to three significant figures

(iv) Calculate the area of $\triangle BAC$.

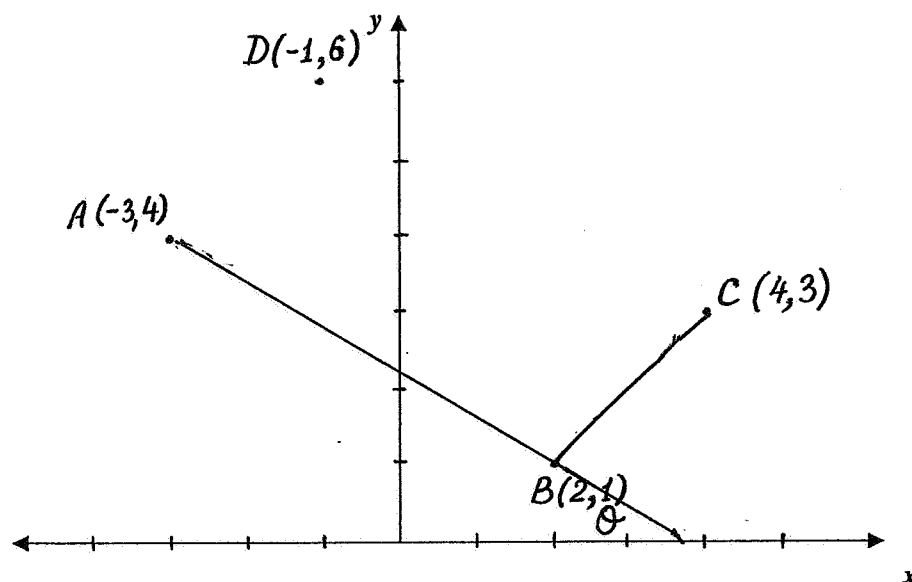
Question 4 – (15 marks)

Marks

- a) Solve simultaneously $2x - y = -1$ and $x + y = 4$ 2
- b) Find the centre and radius of the circle by completing the squares of $x^2 - 4x + y^2 - 5 = 0$ 3
- c) Solve for θ where $0^\circ \leq \theta \leq 360^\circ$: $\tan 2\theta = -\sqrt{3}$ 3
- d) For a certain arithmetic sequence the third term is 3 and the tenth term is 38. Determine the common difference. 3
- e) If $f(x) = x^2 + 2x$ 4
- (i) Find $f(3+h)$ in its simplest form.
- (ii) Evaluate $\lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h}$

Question 5 – (15 marks)

Marks



$ABCD$ is a parallelogram.

- Find the length of BC . 2
- Show that the equation of the line BC is $x - y - 1 = 0$. 3
- Show that AB is parallel to DC . 2
- Find the perpendicular distance from A to BC . 2
- Calculate the area of the parallelogram $ABCD$. 1
- What is the equation of the line through A perpendicular to BC ? 3
- The line through the points A and B makes an angle θ with the positive x -axis. Find θ to the nearest minute. 2

Question 6 – (15 marks)

Marks

- a) At the point (1, 2) on the curve $y = x^3 - 3x + 2$, the value of $\frac{dy}{dx}$ is zero.
What does this tell us about the tangent to the curve at this point? 1
- b) Sketch the function $y = x(x-1)^2$ showing the x and y intercepts. 2
- c) Solve for x :
- (i) $8^x = \frac{1}{16}$ 2
- (ii) $x = \log_4 16$ 1
- (iii) $x = \log_5 20$ to 3 decimal places 2
- d) Consider the series $6 + 12 + 24 + \dots$ 4
- (i) Find S_n , the sum of the first n terms in simplest form.
- (ii) How many terms are required for the sum to be 3066?
- e) Using the sum of an infinite series express $0.\dot{1}2$ as a fraction in its simplest form. 3

Question 7 – (15 marks)

Marks

- a) Sketch any curve for which $\frac{dy}{dx}$ is positive. 1
- b) The lines $3x + 5y - 9 = 0$ and $2x + y + 1 = 0$ intersect at point B . Find the equation of the line through B and point $A(4, -2)$. 3
- c) A circle has a diameter with endpoints $A(2, 4)$ and $B(8, 12)$. Show that $C(1, 11)$ lies on the circle. 3
- d) If $\log_a 5 = m$ and $\log_a 2 = n$ find expressions, in simplest form, for: 3
- i) $\log_a 10$
- ii) $\log_a \left(\frac{2}{5}\right)$
- iii) $\log_a \sqrt{2}$
- e) Calculate x if $(2x - 3)$, $(x + 1)$, $(2 - x)$ are the first three terms of an Arithmetic sequence. 2
- f) Evaluate $\sum_{n=1}^{\infty} 16 \left(\frac{1}{2}\right)^n$ 3

Question 8 – (15 marks)

Marks

- a) For the function $f(x) = x(3x - 1)^4$ 3
- (i) Find $f'(x)$
- (ii) Evaluate $f'(1)$
- b) Sketch the region bounded by $y > x - 2$ and $y \leq 4 - x^2$ 3
- c) Calculate the value of k if the interval joining the points $(1, k - 1)$ to $(k, 2k - 3)$ has a gradient of $\frac{3}{4}$. 2
- d) On a number plane the points $O(0, 0)$, $P(8, 0)$, $Q(10, 5)$ and $R(2, 5)$ form a parallelogram. Show that the diagonals of the parallelogram bisect each other. 3
- e) The numbers x , y and 9 are in arithmetic progression and x , y and 12 are in geometric progression. Hence find the values of x and y . 4

End of Paper

August 2008

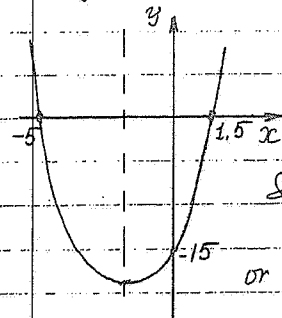
Question 1

a) $\sqrt{\frac{102.743}{449.68}} = 0.477996...$
 $= 0.478$ (1)

b) $(3-2x)^2 - 4(3x+5)$ (1)
 $= 9 - 2 \times 3 \times 2x + 4x^2 - 12x - 20$
 $= 4x^2 - 24x - 11$ (1)

c) $2x^2 - 15 \geq -7x$
 $2x^2 + 7x - 15 \geq 0$
 $2x^2 + 10x - 3x - 15 \geq 0$
 $2x(x+5) - 3(x+5) \geq 0$
 $(2x-3)(x+5) \geq 0$
 $\therefore$ zeros: $x = 1.5$ or $x = -5$ (1)

vertex: $x = \frac{1.5 - 5}{2}$
 $x = -1.75$
 $y = -21.125$
y-intercept is -15



Solution:
 $x \leq -5$ (1)
or $x \geq 1.5$

d) $\frac{7+4\sqrt{3}}{7-4\sqrt{3}} \times \frac{7+4\sqrt{3}}{7+4\sqrt{3}}$ (1)
 $= \frac{(7+4\sqrt{3})^2}{49 - 16 \times 3} = \frac{49 + 2 \times 7 \times 4\sqrt{3} + 48}{1}$
 $= 97 + 56\sqrt{3}$ (1)

e) $\cot \theta \times \sin \theta$

$= \frac{\cos \theta}{\sin \theta} \times \sin \theta = \cos \theta$ (1)

f) $\operatorname{cosec} 240^\circ = \frac{1}{\sin 240^\circ}$ (1)

$= \frac{1}{\sin (180^\circ + 60^\circ)}$

$= \frac{1}{-\sin 60^\circ}$

$= -\frac{1}{\frac{\sqrt{3}}{2}}$

$= -\frac{2}{\sqrt{3}}$ (1)

g) (i) $\frac{dy}{dx} = (2x^{\frac{1}{2}})' = x^{-\frac{1}{2}}$
 $= \frac{1}{\sqrt{x}}$ (1)

(ii) Let $u = 3x - 4$
then $y = u^4$
 $\frac{dy}{du} = 4u^3 = 4(3x-4)^3$ (1)
 $\frac{du}{dx} = 3$

$\frac{dy}{dx} = 12(3x-4)^3$ (1)

or using powers of a linear function formula (1)

$\frac{dy}{dx} = 4 \times 3 \times (3x-4)^3 = 12(3x-4)^3$ (1)

(iii) Let $u = 2x - 1$ and $v = 2x + 1$
then $\frac{du}{dx} = 2$ and $\frac{dv}{dx} = 2$ (1)

Using the quotient rule:

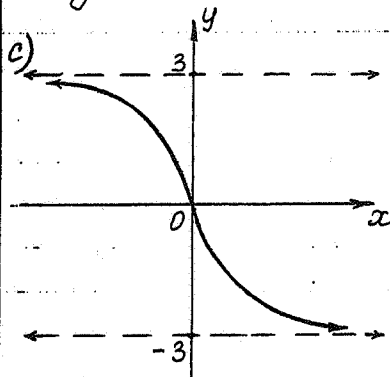
$\frac{dy}{dx} = \frac{(2x+1) \times 2 - (2x-1) \times 2}{(2x+1)^2} = \frac{4x+2-4x+2}{(2x+1)^2}$
 $= \frac{4}{(2x+1)^2}$ (1)

End of Question 1

Question 2

$$\begin{aligned} \text{a) } \frac{2x}{x-3} + \frac{3x}{x-2} &= \frac{2x(x-2) + 3x(x-3)}{(x-3)(x-2)} \quad (1) \\ &= \frac{2x^2 - 4x + 3x^2 - 9x}{(x-3)(x-2)} \\ &= \frac{5x^2 - 13x}{(x-3)(x-2)} \quad (1) \\ \text{or } &= \frac{5x^2 - 13x}{x^2 - 5x + 6} \end{aligned}$$

$$\begin{aligned} \text{b) } f'(x) &= 6x - 2 \\ f'(-1) &= -6 - 2 = -8 \quad (1) \\ \text{Using point-gradient formula} & \\ m &= -8 \quad P(-1, 6) \\ y - 6 &= -8(x + 1) \quad (1) \\ y &= -8x - 8 + 6 \\ y &= -8x - 2 \quad (1) \end{aligned}$$



- (1) For the correct curve
(1) For the asymptote $y = -3$
Domain: all real x (1)
Range: $-3 < y < 3$ (1)

$$\text{d) } (i) \tan \theta = \frac{126.5}{83.9} \quad (1)$$

$$\theta = \tan^{-1}\left(\frac{126.5}{83.9}\right)$$

$$\theta = 56.4460267^\circ$$

$$\approx 56^\circ 27' \quad (1)$$

$$(ii) \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\cos \theta = \frac{55.8^2 + 44.3^2 - 18.3^2}{2 \times 55.8 \times 44.3} \quad (1)$$

$$= \frac{4741.24}{4943.88}$$

$$\theta = \cos^{-1}\left(\frac{4741.24}{4943.88}\right)$$

$$= 16.461179^\circ$$

$$\approx 16^\circ 28' \quad (1)$$

$$\begin{aligned} \text{e) } 14x^{-2}y^{-1} \div 7^2xy^{-3} &= \frac{14}{x^2y} \div \frac{7^2x}{y^3} \quad (1) \\ &= \frac{14}{x^2y} \times \frac{y^3}{49x} = \frac{2y^2}{7x^3} \quad (1) \end{aligned}$$

End of question 2

Question 3

a)

$$(1) \text{ gradient of } AB = \frac{5+1}{3-1} = \frac{6}{2} = 3$$

$$(1) \text{ gradient of } BC = \frac{8-5}{4-3} = \frac{3}{1} = 3$$

$$\text{gradient of } AB = \text{gradient of } BC$$

Point B is common

$\therefore$ Points A(1, -1), B(3, 5) & C(4, 8) are collinear. (1)

$$\text{b) } 8, 11, 14 \dots$$

$$(i) a = 8 \quad d = 3$$

$$T_n = a + (n-1)d$$

$$T_n = 8 + 3(n-1)$$

$$T_n = 8 + 3n - 3$$

$$= 3n + 5 \quad (1)$$

(ii)

$$3n + 5 > 1000 \quad (1)$$

$$3n > 1000 - 5$$

$$n > 995 \div 3$$

$$n > 331.67$$

The first term which

is greater than 1000 (1)

is the 332nd term

and its value is

$$T_{332} = 8 + 331 \times 3$$

$$= 1001$$

(iii)

$$S_n = \frac{1}{2}n(2a + (n-1)d) \quad (1)$$

$$S_{20} = \frac{1}{2} \times 20(2 \times 8 + 19 \times 3)$$

$$= 10 \times 38 = 380 \quad (1)$$

c)

$$(i) \sin 40^\circ = \frac{250}{AC} \quad (\text{opposite} / \text{hypotenuse})$$

$$\therefore AC = \frac{250}{\sin 40^\circ} \quad (1)$$

(ii) In the $\triangle ABC$

$$\angle ACB = 180^\circ - 70^\circ = 110^\circ$$

(angles on a straight line add up to 180°) (1)

$$\angle BAC = 180^\circ - 110^\circ - 40^\circ$$

$$= 30^\circ$$

(Angle sum of a triangle is 180°) (1)

(iii)

$$\tan 40^\circ = \frac{250}{BD}$$

$$BD = \frac{250}{\tan 40^\circ} \quad (1)$$

$$\tan 40^\circ = \frac{250}{CD}$$

$$CD = \frac{250}{\tan 70^\circ} \quad (1)$$

$$BC = BD - CD$$

$$= \frac{250}{\tan 40^\circ} - \frac{250}{\tan 70^\circ} = 206.9458 \dots$$

$$\approx 207 \text{ m} \quad (1)$$

$$(iv) \text{ Area} = \frac{1}{2}bc \sin A$$

$$\text{Area} = \frac{1}{2} AC \cdot BC \cdot \sin \angle ACB$$

$$= \frac{1}{2} \times \frac{250}{\sin 70^\circ} \times 207 \times \sin 110^\circ \quad (1)$$

$$= \frac{1}{2} \times 266 \times 207 \times \sin 110^\circ$$

$$= 25874.6775 \dots \quad (1)$$

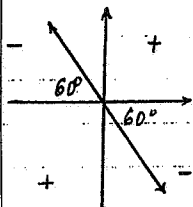
$$\approx 25871 \text{ m}^2 \text{ (to the nearest m)}$$

Question 4

a) $2x - y = -1$ (1)
 $x + y = 4$ (2)
 (1) + (2)
 $3x = 3$
 $x = 1$ (1)
 Substitute x into (2)
 $1 + y = 4$
 $y = 3$ (1)
 $\therefore x = 1$ and $y = 3$

b) $x^2 - 4x + y^2 - 5 = 0$
 $x^2 - 4x + 4 + y^2 = 5 + 4$
 $(x - 2)^2 + y^2 = 9$ (1)
 Centre $O(2, 0)$ (1)
 Radius $r = \sqrt{9} = 3$ (1)

c) $\tan 2\theta = -\sqrt{3}$ $0^\circ \leq \theta \leq 360^\circ$
 Let $u = 2\theta$
 Then $\tan u = -\sqrt{3}$
 The restriction on u :
 $0^\circ \leq u \leq 720^\circ$ (1)



From the diagram: (1)
 $u = 120^\circ, 300^\circ, 480^\circ$ or 660°
 $\theta = \frac{u}{2} = 60^\circ, 150^\circ, 240^\circ$ or 330° (1)

d)

$T_n = a + (n-1)d$
 $T_3 = a + 2d = 3$ (1)
 $T_{10} = a + 9d = 38$ (1)
 Two simultaneous equations:
 $a + 2d = 3$ (1)
 $a + 9d = 38$ (2)
 Subtract (1) from (2):
 $7d = 35$
 $d = 5$ (1)

e) $f(x) = x^2 + 2x$
 (i) $f(3+h) = (3+h)^2 + 2(3+h)$
 $= 3^2 + 2h \cdot 3 + h^2 + 2 \cdot 3 + 2h$
 $= 15 + 8h + h^2$ (1)

(ii) $\lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} =$
 $\lim_{h \rightarrow 0} \frac{15 + 8h + h^2 - (9 + 6)}{h}$ (1)
 $= \lim_{h \rightarrow 0} \frac{8h + h^2}{h} = \lim_{h \rightarrow 0} \frac{h(8+h)}{h}$
 $= \lim_{h \rightarrow 0} (8+h) = 8$ (1)

end of question 4

Question 5

a) $BC = \sqrt{(3-1)^2 + (4-2)^2}$ (1)
 $= \sqrt{4+4}$
 $= \sqrt{8} = 2\sqrt{2}$ (1)

b) $m = \frac{3-1}{4-2} = 1$ (1)
 Using point-gradient formula:
 $m = 1$, $B(2, 1)$
 $(y-1) = 1(x-2)$ (1)
 $y = x - 1$
 $-x + y + 1 = 0$
 $x - y - 1 = 0$ (1)

c) gradient of $AB = \frac{1-4}{2+3} = \frac{-3}{5}$ (1)
 gradient of $DC = \frac{3-6}{4+1} = \frac{-3}{5}$ (1)
 $\therefore AB \parallel DC$ (as their gradients are equal) (1)

d) $A(-3, 4)$
 $BC: x - y - 1 = 0$ (1)
 $p = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$ (1)
 $p = \frac{|1(-3) - 4 - 1|}{\sqrt{1^2 + (-1)^2}} = \frac{8}{\sqrt{2}}$
 $= \frac{8\sqrt{2}}{2} = 4\sqrt{2}$

e) $A = BC \times h$
 $A = 2\sqrt{2} \times 4\sqrt{2}$
 $A = 16 \text{ units}^2$ (1)

f) gradient $= -\frac{1}{\text{gradient } BC}$

$m = -\frac{1}{1} = -1$ (1)

$A(-3, 4)$

Using point-gradient formula:

$y - 4 = -1(x + 3)$ (1)

$y - 4 = -x - 3$

$x + y + 1 = 0$ (1)

g) gradient $AB = \frac{-3}{5}$ (from part c)

$\tan \theta = -\frac{3}{5}$ (1)

$\theta = 180 - \tan^{-1}\left(\frac{3}{5}\right)$

$\theta = 180^\circ - 30^\circ 58'$

$= 149^\circ 2'$ (1)

End of question 5

Question 6

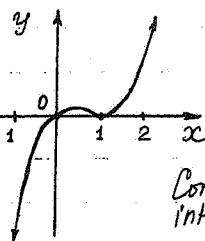
a) The tangent to the curve at this point is horizontal. (1)

b) $y = x(x-1)^2$
x-intercepts: $x=0$ and $x=1$ (zeros)

y-intercept: $y=0$

Sign of the function near zeros:

x	-1	0	$\frac{1}{2}$	1	2
y	-4	0	$\frac{1}{8}$	0	2
Sign	-	0	+	0	+



Correct intercepts (1)

Correct direction (1)

c) (i) $8^x = \frac{1}{2^{3x}} = \left(\frac{1}{2}\right)^{3x}$ (1)

$3x = -4$
 $x = -\frac{4}{3}$ (1)

(ii) $x = \log_4 16$
 $x = \log_4 4^2$
 $= 2 \log_4 4 = 2$ (1)

(iii) $x = \log_5 20$
 $= \frac{\log_{10} 20}{\log_{10} 5} = 1.86135...$
 ≈ 1.861
(Correct to 3 d.p.) (1)

d) (i) $S_n = \frac{a(r^n-1)}{r-1}$ (1)

$a=6, r=2$
 $\therefore S_n = \frac{6(2^n-1)}{2-1}$

$= 6(2^n-1)$ (1)

(ii) $6(2^n-1) = 3066$ (1)

$2^n-1 = 511$

$2^n = 512$

$n = \log_2 512$

$n = \frac{\log_{10} 512}{\log_{10} 2}$

$n = 9$ (1)

$\therefore$ 9 terms are required for the sum to be 3066.

e) $0.12 = 0.1 + 0.02 + 0.002 + \dots$ (1)
GP with $a=0.02$
and $r=0.1$

$0.12 = 0.1 + \frac{a}{1-r}$ (1)

$= 0.1 + \frac{0.02}{0.9}$

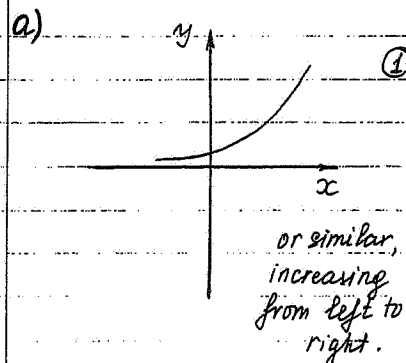
$= \frac{1}{10} + \frac{2}{90}$

$= \frac{9}{90} + \frac{2}{90} = \frac{11}{90}$ (1)

End of question 6

Question 7

Marks



b) $3x + 5y - 9 + k(2x + y + 1) = 0$ (1)

Substituting $A(4, -2)$:

$3 \times 4 + 5 \times (-2) - 9 + k(2 \times 4 - 2 + 1) = 0$

$12 - 10 - 9 + 7k = 0$

$7k = 7$

$k = 1$ (1)

Substituting k :

$3x + 5y - 9 + 1(2x + y + 1) = 0$

$5x + 6y - 8 = 0$ (1)

c) $AB = \sqrt{(8-2)^2 + (12-4)^2} = \sqrt{100} = 10$ units
AB is the diameter

$\therefore$ the radius = 5 units (1)

Let Q be the centre of the circle

$Q\left(\frac{8+2}{2}, \frac{12+4}{2}\right)$

$Q(5, 8)$ (1)

$QC = \sqrt{(1-5)^2 + (11-8)^2} = 5$ units

QC = the radius

$\therefore$ Point C(1, 11) lies on the circle. (1)

d)

(i) $\log_a 10 = \log_a 2 \times 5$
 $= \log_a 2 + \log_a 5$
 $= n + m$ (1)

(ii) $\log_a \left(\frac{2}{5}\right) = \log_a 2 - \log_a 5$
 $= n - m$ (1)

(iii) $\log_a \sqrt{2} = \log_a 2^{\frac{1}{2}}$
 $= \frac{1}{2} \log_a 2$
 $= \frac{n}{2}$ (1)

e) common difference = $T_2 - T_1$
 $d = (x+1) - (2x-3)$
 $= x - 2x + 1 + 3 = -x + 4$
 $= 4 - x$ (1)

$d = T_3 - T_2$
 $(2-x) - (x+1) = 4 - x$
 $-2x + 1 = 4 - x$
 $-x = 3$
 $x = -3$ (1)

f) $\sum_{n=1}^{\infty} 16 \times \left(\frac{1}{2}\right)^n = 8 + 4 + 2 + \dots$ (1)

$a = 8, r = \frac{1}{2}$
This is a geometric series with
 $\therefore \sum_{n=1}^{\infty} 16 \times \left(\frac{1}{2}\right)^n = S_{\infty} = \frac{a}{1-r}$ (1)
 $= \frac{8}{1-\frac{1}{2}}$
 $= 8 \times 2$
 $= 16$ (1)

End of question 7

Question 8

a)

(i) Let $y = x(3x-1)^4$,

$$u = x$$

$$v = (3x-1)^4$$

Then: $y = u \times v$

$$u' = \frac{du}{dx} = 1$$

$$v' = \frac{dv}{dx} = 4 \times 3(3x-1)^3$$

$$= 12(3x-1)^3 \quad (1)$$

Using the product rule:

$$= (3x-1)^4 \times 1 + x \times 12(3x-1)^3$$

$$= (3x-1)^4 + 12x(3x-1)^3$$

$$= (3x-1)^3(3x-1+12x)$$

$$= (3x-1)^3(15x-1) \quad (1)$$

(ii)

$$f'(1) = (3-1)^3(15-1)$$

$$= 8 \times 14 = 112 \quad (1)$$

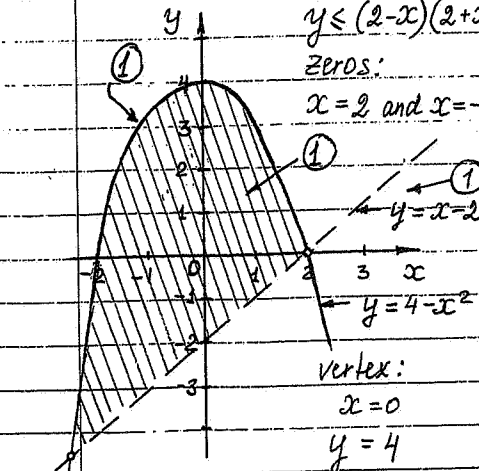
b)

$$y > x-2 \quad y \leq 4-x^2$$

$$y \leq (2-x)(2+x)$$

Zeros:

$$x=2 \text{ and } x=-2$$

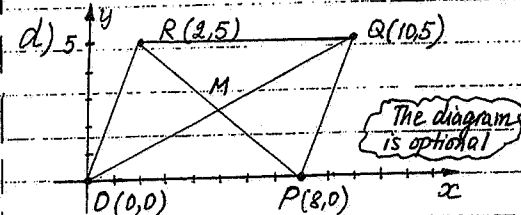


c) $\frac{(2k-3)-(k-1)}{k-1} = \frac{3}{4} \quad (1)$

$$\frac{2k-3-k+1}{k-1} = \frac{3}{4}$$

$$\frac{k-2}{k-1} = \frac{3}{4}$$

$$k=5 \quad (1)$$



Let M be the point of intersection of the diagonals OQ and PR (1)

$$\text{Midpoint of } OQ = \left(\frac{0+10}{2}, \frac{0+5}{2}\right) = (5, 2.5)$$

$$\text{Midpoint of } PR = \left(\frac{8+2}{2}, \frac{0+5}{2}\right) = (5, 2.5)$$

Since two different straight lines can only have one common point, (1)

$$\text{Midpoint of } OQ = \text{Midpoint of } PR = M \quad (1)$$

e)

Since $x, y, 9$ form an AP: $y-x = 9-y$
 $-x = 9-2y$
 $x = 2y-9 \quad (1)$

Since $x, y, 12$ form a GP: $\frac{y}{x} = \frac{12}{y}$
 $y^2 = 12x \quad (2)$

Substitute (1) into (2):

$$y^2 = 12(2y-9)$$

$$y^2 - 24y + 108 = 0$$

$$y^2 - 18y - 6y + 108 = 0$$

$$y(y-18) - 6(y-18) = 0$$

$$(y-6)(y-18) = 0$$

$$y=6 \text{ or } y=18 \quad (1)$$

Substitute y into (1)

$$y=6 \quad x=3 \text{ or } y=18 \quad x=27$$

End of Question 8